Lab Assignments: Inline Functions and Function Overloading

Department: Computer Science and Engineering, IUST Awantipora
Course: Object-Oriented Programming with C++

Assignment 1: Square and Cube

Write inline functions **int square(int n)** and **int cube(int n)**. Print a table of n, square and cube for n = 1 to 5.

Output: 1 1 1 | 2 4 8 | 3 9 27 | 4 16 64 | 5 25 125

Assignment 2: Largest and Smallest

Write inline functions **int maxOf(int a, int b)** and **int minOf(int a, int b)**. Use them to find the largest and smallest of three numbers.

Input: 12 45 7
Output: Max = 45, Min = 7

Assignment 3: Leap Year and Even Number

Write inline functions **bool isLeap(int year)** and **bool isEven(int n)**.

Input: 2024 | 1900 | 2000
Output: Leap | Not leap | Leap

Assignment 4: Temperature Converter

Write inline functions **double cToF(double c)** and **double fToC(double f)**. Drive them from a menu.

Input: 37 C | 212 F
Output: 98.6 F | 100 C

Assignment 5: Inline Function vs Macro

Define **#define SQR(x) x*x** and **inline int sqr(int x)**. Print **SQR(2+3)** and **sqr(2+3)** and explain the difference.

Output: SQR(2+3) = 11 | sqr(2+3) = 25

Assignment 6: Area of Shapes

Overload **area()** for a circle **(double r)**, a rectangle **(double l, double b)** and a triangle **(double a, double b, double c)** using Heron's formula.

Input: r = 2 | 4 5 | 3 4 5
Output: 12.5664 | 20 | 6

Assignment 7: Overloaded add()

Overload **add()** for **(int, int)**, **(double, double)**, **(int, int, int)** and **(string, string)**.

Input: 3 4 | 2.5 1.25 | 1 2 3 | "Hello, " "IUST"
Output: 7 | 3.75 | 6 | Hello, IUST

Assignment 8: Overloaded print()

Overload **print()** for **int**, **double**, **char** and **string**. Each version prints the type name and the value.

Input: 10 | 3.14 | 'A' | "IUST"
Output: int: 10 | double: 3.14 | char: A | string: IUST

Assignment 9: Overloaded swap

Overload **swapValues()** for **int&**, **double&** and **char&** parameters.

Input: 1 2 | 1.5 2.5 | 'X' 'Y'
Output: 2 1 | 2.5 1.5 | Y X

Assignment 10: Overloaded Inline maxOf()

Write overloaded inline **maxOf()** for **(int, int)**, **(double, double)**, **(int, int, int)** and **(const int *arr, int n)**. Then call **maxOf(5, 2.5)** and note the compiler error.

Input: 3 8 | 2.5 1.5 | 4 9 6 | {4, 19, 7, 11}
Output: 8 | 2.5 | 9 | 19
Error: call of overloaded 'maxOf(int, double)' is ambiguous